Simplifying GBS circuits before running them on gate-based hardware

1. Optical level (Blackbird, before compilation)

1.1 Exact

#	Rule	Why it's exact	Status
E1	Drop `BSgate(θ, φ)` with θ ≡ 0 (mod 2π)	BS(0, φ) = identity for any φ	
E2	Drop `Rgate(φ)` with φ ≡ 0 (mod 2π)	identity	
E3	Merge consecutive `Rgate`s on one mode, then apply E2	R(a)·R(b) = R(a+b)	
E4	Drop `Sgate` with r numerically 0	S(0) = identity. Happens for rank-deficient adjacency matrices (zero Takagi values)	
E5	Track vacuum modes: a mode is vacuum until squeezed or mixed with a non-vacuum mode. Drop `Rgate` on a vacuum mode	R(φ)|0⟩ = |0⟩	
E6	Drop `BSgate` between two vacuum modes	a passive transform maps vacuum to vacuum	
E7	Drop `Rgate`s after a mode's last `BSgate`	phase commutes with photon-number measurement	
E8	Replace `BSgate(θ ≡ π)` by `Rgate(π)` on both modes (then E3/E7 usually remove them)	BS(π, φ): a → −a, b → −b for any φ	saves 28 CNOT each; rare
E9	Replace `BSgate(θ ≡ π/2)` by a mode relabelling plus phases, carried through the rest of the circuit; un-permute the sampled modes afterwards	BS(π/2, φ) is a mode swap up to phases	saves 28 CNOT each; changes the sampler's output order, so it must return the permutation
E10	Merge consecutive `BSgate`s on the same mode pair (with `Rgate`s between) into one `BSgate` plus `Rgate`s	every 2-mode passive transform equals R·BS·R	Clements meshes have no such pairs (N(N−1)/2 is already minimal), so this only helps other circuit generators
E11	Absorb an `Rgate` directly after an `Sgate` on a fresh vacuum mode into the squeezing phase	R(φ)·S(r, θ)|0⟩ = S(r, θ - 2φ)|0⟩	0 CNOT gain
E12	Remove modes that stay vacuum for the whole circuit from the	they always measure 0 photons	saves 2 qubits per mode

#	Rule	Why it's exact	Status
	program, and re-insert zero columns into the samples		

1.2 Approximate

#	Rule	Error control
A1	Drop `Sgate` with r < `r_threshold`	Mean photon number removed = Σ sinh²(r). Combined with E5/E6, whole vacuum regions of the mesh disappear
A2	Drop `BSgate` with small |θ| (not just numerically zero)	error grows with θ and with the photon number in the two modes
A3	Replace `BSgate` with θ near π/2 by a swap (then E9)	as A2, around π/2
A4	Low-rank approximation of the adjacency matrix: keep the k largest Takagi values, giving fewer squeezers and, with E12, fewer beam splitters; basically equivalent of A1	discarded Takagi weight
A5	Lower `n_mean_photons` so fewer high photon-number states are populated	changes the GBS distribution itself

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2. Compiler level (inside `gbs-compiler`)

#	Idea	Exact?	Notes
C1	Drop the higher photon-number sector layers of the BS decomposition	approximate	the cost/error trade-off can be computed exactly per sector
C2	Hardware-aware BS decomposition	exact	choose layout of two-qubits gate in BS according to hardware topology
C3	Stop emitting rotations by 0 (e.g. `rz(-0.0)`	exact	the transpiler removes them anyway; mostly cosmetic
C4	Other encodings: Gray or unary instead of binary trade qubits for cheaper number-conserving gates	exact	research direction
C5	Squeezing at the begining could be reduced to single qubit gate $S(z)\|0\rangle = (R_z(\phi-\pi)R_y(\theta)) \otimes \mathbb{I}\|0\rangle$	exact	only for preparing squeezed states from vacuum, which is the GBS case

#	Idea	Exact?	Notes
C6	BS directly following prepared squeezed states can be reduced, because they have as inputs only states with 0 or 2 photons	exact	only for first layer of BS in GBS